Health-a-thon 2026

ROUND 2

SHISHU

A multilingual WhatsApp & voice companion that guides mothers for further evaluation to recognize inherited disorders in their children — early enough to matter.

TEAM: OPFORD / SHISHU **TEAM CODE: FHUYZI**

Team overview and background

OPFORD
OPEN PLATFORM FOR ORPHAN DISEASES

A not-for-profit improving outcomes in rare and neglected inherited disorders through credible information, counselling and collaboration.
opford.org

ARTPARK
AI & Robotics Technology Park @ IISc

Brings together research, technology and industry expertise to develop AI and robotics solutions for societal impact.
artpark.in

A collaboration formed for Health-a-thon 2026

The SHISHU idea and collaboration began during the challenge, combining clinical, counselling and AI expertise.

Core team

Dr Poulami Das
Doctor - Pediatric Hematologist Oncologist

Deeksha
Technology, AI

Meher Zaheer
Coordination

Prof. Vijay Chandru
Strategy

Advisors

Jigar Doshi	AI
Venu Nair	Communications
Dr Dipanjana Datta	Genetics
Lakshmi	Counselling

Problem statement and supporting evidence

Inherited diseases and diagnostic odyssey

70 million Indians live with a rare disease, four in five of them genetic

Hemoglobinopathies, inborn errors of metabolism and of immunity, neuromuscular disorders and chromosomal syndromes make up most of the burden.

Endogamy and consanguinity concentrate the risk

Marriage within a community, sustained across generations, drives both the incidence and the prevalence of these disorders in India.

The warning signs sit across a family tree

Faltering growth, lost milestones, recurrent infections and repeated infant deaths accumulate over years, and across a pedigree nobody ever draws.

A doctor’s visit is a snapshot

Records do not follow the family and genetic tests are rarely ordered.

Most families are unaware of this risk, and some families get to know too late

After years of illness, irreversible damage, or another affected birth — by which time sometimes, a diagnosis can no longer change the outcome, only explain it.

Target users and use case

A mother's account of her family's health, ready for a more informed care conversation

PRIMARY USER

Mothers

Caregivers share concerns about a child, pregnancy or family history in everyday language.

SETTING AND ACCESS

Introduced through trusted care channels

A doctor, clinic, ASHA or ANM can introduce SHISHU. A caregiver can also open a link independently.

At home, before an appointment or between visits

The challenges it addresses

Limited time for detailed history

ASHAs cover scattered households and have little time beyond their paperwork.

A visit is only a snapshot

Records do not follow the family, so relevant history can be missed across visits.

Sensitive facts remain unspoken

Fear of blame can limit disclosure of infant deaths or marriage within the family.

The workflow it supports

1 Doctor shares the link

The doctor or healthcare worker introduces SHISHU and sends the mother the web link.

2 Mother has the conversation

She talks with SHISHU about her child, pregnancy and family health.

3 Mother shares the summary

She takes the PDF summary back to the doctor to discuss her history and concerns.

Solution overview, key features and impact

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A voice conversation in a web app turns family history into a shareable summary.

CURRENT VERSION

1 **Browser-based voice**
A caregiver opens the web app and speaks with SHISHU.

2 **History and follow-up**
Open questions capture the story, then focused questions fill gaps.

3 **Red flags and urgent concerns**
The prototype highlights concerns and can interrupt for emergency advice.

4 **A summary to keep and share**
A PDF captures the conversation for discussion with a healthcare professional.

POTENTIAL IMPACT

What changes for families

1 **A diagnosis that arrives early enough to matter**

2 **The next pregnancy becomes a choice instead of a repeat**

3 **Mothers stop carrying the blame**

SHISHU — System Architecture

CURRENT LIVE DEPLOYMENT

Real-time vernacular Hindi triage voice agent → two-sided post-call intelligence for Mother & Clinician

Backend: FastAPI on Railway (Docker + Persistent Volume at `/data`)
Live Voice Engine: Gemini 3.1 Flash Live · **Post-Call Audit:** Gemini 3.8 Flash

CALLER & BENEFICIARY

Mother

Speaks everyday Hindi in her own words. Shares child health, NICU stay, milestones, antenatal & family history.

◄=► Two-way Hindi Voice Call (WebAudio)

OUTPUT 1 · USER-OWNED ARTIFACT

Mother Hindi PDF (Report)

- ✓ **Actionable Questions:** 3–6 tailored questions for next doctor or ASHA/ANM visit (empowerment, not passive summary).
- ✓ **Positive Disclosures:** Clear Devanagari sentences giving exact context of reported concerns.
- ✓ **Pertinent Negatives:** Structured checklist of screened topics confirmed normal.
- ✓ **Antenatal Guidance:** Universal thalassemia carrier test prompt before next pregnancy.

▲ Delivered directly to Mother (in-browser download)

INTEGRATED PLATFORM ENGINE

FASTAPI ON RAILWAY · VOLUME AT /DATA

WebSocket PCM 16kHz in / 24kHz out · SQLite WAL · Session Auth

Web Browser Client

FRONTEND INTERFACE

- **Interactive Voice Call:** One-tap mic, WebAudio streaming.
- **Live Captions:** Instant Hindi speech-to-text / text-to-speech.
- **Client-Side VAD:** 750ms silence detection + 300ms prefix padding.
- **Instant PDF Delivery:** Rendered on-demand at call completion.

Gemini 3.1 Flash Live Pipeline

LIVE SPEECH ENGINE

- **Bilingual Voice:** Warm female voice (Kore) speaking simple Hindi; never changes register.
- **Turn-by-Turn Logging:** Invokes `record_finding` on each answer to record topic & flag status.
- **Safety Limits:** 30s silence auto-wrap with polite Hindi cue; 20-minute ceiling.
- **Dual-Engine Architecture:** Gemini 3.1 Flash Live default; OpenAI Realtime standby.

CLINICAL DIALOGUE INTELLIGENCE & SAFETY RULES (IN PROMPTS & STATE MACHINE)

LOW COGNITIVE LOAD · CULTURALLY GROUNDED

🗣️ Vernacular Clinical Phrasing
Everyday non-stigmatizing Hindi: **"kaanch ka box"** (NICU), **"neeli light"** (phototherapy), **"naal 3 hafte se zyada"** (delayed cord separation), **"bagal mein gaanth"** (BCG reaction), **"jhatke"** (fits), **"seekha hua bhool jaana"** (neuroregression), **"tilli/pet badhna"** (organomegaly), **"neel padna"** (bruising).

🎯 Single-Turn & Wide-First
Strictly one short question per turn. Never bundles multiple inquiries. Asks **one wide question first** (e.g. pregnancy complications); only drills down into details (who, when, severity) one by one after an affirmative answer.

🧬 Adaptive Demographic Gating
Dynamic branching: multi-child scope (**"aapke sabhi bachchon mein se..."**); puberty asked only if child ≥10y; periods only if daughter ≥11y; milk teeth only if child ≥5y; unmarried/no-child callers skip pregnancy and focus on thalassemia & family pedigree.

STRICT GUARDRAILS

- **No-Diagnosis Hard Rule:** Never utters disease names to caller.
- **Emergency Halt:** Active danger signs trigger immediate 108 referral & call stop.
- **Non-Judgmental Tone:** Respectful on high parity (5+ kids), loss, or consanguinity.
- **Privacy Storage:** Encrypted SQLite on volume; no unauthorized EHR export.

Topic Coverage Auditor

GEMINI 3.8 FLASH

- Audits finished transcript against 40+ triage topics.
- Categorizes answers: `yes`, `no`, `unknown`, or `not_asked`.
- Distinguishes clinical flags from normal developmental variants.
- Configurable via DB settings (fallback to `gpt-4o-mini`).

12 NRROID Gates & Note Synthesis

CLINICAL SYNTHESIS

- Cross-references National Policy for Rare Diseases screening criteria.
- Evaluates infant deaths, consanguinity, and familial early-onset risk.
- Distinguishes 1st-degree relatives from distant relatives for CVD/stroke.
- Renders WeasyPrint PDF layout with Noto Sans Devanagari typography.

CLINICAL CARE PROVIDER

Doctor / Clinician

Medical Specialist, Geneticist, Medical Officer, or ASHA/ANM Supervisor.

==► Delivered via Doctor / Admin Portal

OUTPUT 2 · PROFESSIONAL TRIAGE NOTE

Clinical Triage Note (English)

- ✓ **8 Standardized Headings:** Profile & Chief Complaint, Positive Red Flags, Neonatal, Development & Milestones, Systemic & Hematology, Antenatal, Family Pedigree, Pertinent Negatives.
- ✓ **NRROID Gate Alerts:** Explicit flagging of rare disease indicators.
- ✓ **Full Audio & Transcript:** Synchronized dual-channel WAV audio playback for clinical verification.
- ✓ **Call Telemetry:** Completion status, duration, silence counts, and emergency halt logs.

🔒 Secured via Clinical / Admin Passcode

Build sprint journey and progress

From a new collaboration in September to a working web app tested on 7 November

Step	Owners	Timeline	Outcome
New team and concept	Team	7–13 Sep	Formed the Health-a-thon collaboration and defined a caregiver assistant focused on family health history.
Phase 1 idea submission	Team	14 Sep–3 Oct	Combined clinical and technical inputs into the SHISHU concept note and submitted the Phase 1 proposal.
Building questionnaire	Dr Poulami + Lakshmi	5–23 Oct	Developed the history-taking questions and approach, covering the child, pregnancy and family history.
Example calls and conversational flow	Dr Poulami + Lakshmi	19 Oct–6 Nov	Recorded role-play conversations and used them to refine open questions, sensitive wording and follow-up prompts.
Voice web app development and testing	Deeksha + Jigar	1–7 Nov	Built and tested the browser voice conversation, history summary and PDF report generation and sharing.

BEFORE GRAND FINALE

Refine and retest the prototype, begin a supervised clinic pilot and review conversations and PDF summaries with clinicians.

Evaluation and user validation

01 Simulated user testing

Human-to-human role-play only, conducted by a paediatric haematologist-oncologist and a counsellor who works with users and patients and understands everyday conversations.

Ongoing internal team review of the bot.

02 Qualitative expert feedback

Several dozen calls by the counsellor and doctor helped improve cultural contextualisation of bot language, easy-to-understand questions and the right cognitive load, avoiding too many or too few questions.

Usability testing and user experience design.

NEXT STEPS

Real-user validation before the grand finale

1 Test with real users

Test mothers in Mumbai and Bangalore clinics and institutes, with a doctor present and experts judging the calls.

2 Apply the evaluation framework

A systematic evaluation metric has been developed.

3 Run accuracy evaluations

Use the real-user tests to evaluate accuracy before the grand finale.

Safety and responsible AI

01 Privacy, consent and security

Obtain informed consent before collecting or recording. Collect only the minimum data needed, restrict access to authorised users, encrypt data, and define retention and deletion. Use synthetic data in public materials.

02 AI limitations and error handling

AI can misunderstand, omit or generate inaccurate content, including mishearing critical voice inputs such as “nahi”. It is not a diagnostic tool. Users can correct factual errors. Urgent or out-of-scope requests are routed to a clinician.

03 Safeguards and human oversight

A clinician reviews the history and proposed summary before it informs care. Set clear clinical boundaries, escalation paths, audit logs and named accountability for decisions and corrections.

04 Bias, accessibility and inclusion

Currently Hindi only, using simple everyday terms such as “kaanch ki box” for NICU. Test across literacy levels and backgrounds to reduce bias. Plan more languages and typed input for users who cannot speak privately.

Deployment, cost and sustainability plan

1 Doctor recommends screening

A few clinics in Mumbai and Bangalore offer mothers a free screening.

2 Mother completes the call

She can use SHISHU in the waiting room or between visits.

3 Mother shares the PDF

She sends the summary to the phone number chosen by the clinic.

What the clinic needs

A link or QR code and one phone number for receiving the PDF.

A short staff briefing. No new clinic software.

Integration

None for this pilot. Later, the same PDF can go into the institute's patient management system.

WHAT THE PILOT COSTS

₹12.4 ± ₹4.3

AI cost per screening

Call length: 9.3 ± 3.4 minutes

₹425 / month

Backend hosting shared by clinics

Up to 1,900 people across all clinics

Monthly clinic cost = screening costs + hosting share

SCALE

At 100,000 people a month, self-hosting and a small language model can bring this to ₹5 ± ₹2 per person.
Self-hosting also keeps the audio and the PDF off a third-party voice API

Next steps and submission decisions

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01 Pilot in clinical practice

Start with a few Mumbai and Bangalore clinics through OPFORD and the clinical team's networks. The doctor shares a link or QR code, the mother completes the conversation, and she returns the PDF to the doctor.

02 Strengthen and validate

Retest red-flag, no-red-flag and emergency scenarios. Review both the conversation and PDF with clinicians. Refine missed context, repeated questions, plain-language prompts and reliable call ending.

03 Close the workflow loop

For the initial pilot, the mother shares the PDF to a clinic-selected number. Explore automatic clinic delivery, patient-record integration and location-specific referral resources as later enhancements.

04 Fund and scale responsibly

Seek a grant to strengthen the technical team, cover infrastructure and support pilot operations. Use pilot feedback and observed usage costs to plan expansion to additional clinics.

PUBLIC SHARING: YES Make solution information available on Health-a-thon and other public websites.

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SHISHU

OPFORD collaboration for the Health-a-thon 2026 Maternal & Child Health track.
Team code: FHUYZI. Contact email: deeksha.j@opford.org

Thank you